Polygons & Quadrilaterals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Proving triangles similar**

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 14 cm. Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

2 **[Foundational]** • **Target: Proving triangles similar**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 11. Substituting $x = 1$ into $y = 3x + 8$ gives $y = 3(1) + 8 = 11$.

3 **[Foundational]** • **Target: Proving triangles similar**

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 90°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 52^\circ) = 90^\circ$.

4 **[Foundational]** • **Target: Proving triangles similar**

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 21 cm. Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

5 **[Foundational]** • **Target: Proving triangles similar**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 14. Substituting $x = 2$ into $y = 3x + 8$ gives $y = 3(2) + 8 = 14$.

6 **[Foundational]** • **Target: Proving triangles similar**

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 83°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 59^\circ) = 83^\circ$.

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7 [Foundational] • Target: Proving triangles similar

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 28 cm. Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Proving triangles similar

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 17. Substituting $x = 3$ into $y = 3x + 8$ gives $y = 3(3) + 8 = 17$.

9 [Intermediate] • Target: Proving triangles similar

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 76°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 66^\circ) = 76^\circ$.

10 [Intermediate] • Target: Proving triangles similar

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 35 cm. Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

11 [Intermediate] • Target: Proving triangles similar

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 20. Substituting $x = 4$ into $y = 3x + 8$ gives $y = 3(4) + 8 = 20$.

12 [Intermediate] • Target: Applications of similar triangles

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 96°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 45^\circ) = 96^\circ$.

13 [Intermediate] • Target: Applications of similar triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

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14 [Intermediate] • Target: Applications of similar triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 13. Substituting $x = 1$ into $y = 4x + 9$ gives $y = 4(1) + 9 = 13$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 10.01 Coordinate Geometry

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 89°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 52^\circ) = 89^\circ$.

16 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

17 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 17. Substituting $x = 2$ into $y = 4x + 9$ gives $y = 4(2) + 9 = 17$.

18 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 3.01 Triangles & Congruence

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 82°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 59^\circ) = 82^\circ$.

19 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 32 cm. Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

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20 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 21. Substituting $x = 3$ into $y = 4x + 9$ gives $y = 4(3) + 9 = 21$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 99° ✓

b. 109°

c. 91°

d. 144°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (36^\circ + 45^\circ) = 99^\circ$.

22 [Foundational] • Target: Review: Similar figures

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

23 [Foundational] • Target: Review: Similar figures

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 16

b. 23

c. 21 ✓

d. 26

Substituting $x = 3$ into $y = 5x + 6$ gives $y = 5(3) + 6 = 21$.

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24 [Foundational] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 78° ✓

b. 88°

c. 70°

d. 144°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (36^\circ + 66^\circ) = 78^\circ$.

25 [Foundational] • Target: Review: Identifying similar triangles

In ΔABC studying review: identifying similar triangles, $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 98° ✓

b. 108°

c. 90°

d. 143°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 45^\circ) = 98^\circ$.

26 [Foundational] • Target: Review: Identifying similar triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

27 [Foundational] • Target: Review: Identifying similar triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 11

b. 15

c. 13 ✓

d. 15

Substituting $x = 3$ into $y = 2x + 7$ gives $y = 2(3) + 7 = 13$.

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28 [Foundational] • Target: Review: Identifying similar triangles

In ΔABC studying review: identifying similar triangles, $m\angle A = 37^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 77° ✓

b. 87°

c. 69°

d. 143°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 66^\circ) = 77^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Review: Similar figures

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

30 [Intermediate] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 92° ✓

b. 102°

c. 84°

d. 144°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (36^\circ + 52^\circ) = 92^\circ$.

31 [Intermediate] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 85° ✓

b. 95°

c. 77°

d. 144°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (36^\circ + 59^\circ) = 85^\circ$.

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32 [Intermediate] • Target: Review: Similar figures

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 21

b. 28

c. 26 ✓

d. 31

Substituting $x = 4$ into $y = 5x + 6$ gives $y = 5(4) + 6 = 26$.

33 [Intermediate] • Target: Review: Identifying similar triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 11

c. 9 ✓

d. 11

Substituting $x = 1$ into $y = 2x + 7$ gives $y = 2(1) + 7 = 9$.

34 [Intermediate] • Target: Review: Identifying similar triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 9

b. 13

c. 11 ✓

d. 13

Substituting $x = 2$ into $y = 2x + 7$ gives $y = 2(2) + 7 = 11$.

35 [Intermediate] • Target: Review: Identifying similar triangles

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

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36 [Intermediate] • Target: Proving triangles similar

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

37 [Intermediate] • Target: Proving triangles similar

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 90° ✓

b. 100°

c. 82°

d. 142°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 52^\circ) = 90^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving triangles similar • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

39 [Challenging] • Target: Proving triangles similar • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 17

b. 22

c. 20 ✓

d. 23

Substituting $x = 4$ into $y = 3x + 8$ gives $y = 3(4) + 8 = 20$.

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40 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 15

c. 13 ✓

d. 17

Substituting $x = 1$ into $y = 4x + 9$ gives $y = 4(1) + 9 = 13$.

41 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 7.01 Circles

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 82° ✓

b. 92°

c. 74°

d. 141°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 59^\circ) = 82^\circ$.

42 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 8.01 Area, Surface Area & Volume

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm ✓

c. 48 cm

d. 200 cm

Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

43 [Challenging] • Target: Perimeters and areas of similar figures • Cross-Lesson: 9.01 Transformations & Symmetry

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

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44 [Challenging] • Target: Perimeters and areas of similar figures • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 8

b. 15

c. 13 ✓

d. 18

Substituting $x = 2$ into $y = 5x + 3$ gives $y = 5(2) + 3 = 13$.

45 [Challenging] • Target: Perimeters and areas of similar figures • Cross-Lesson: 10.01 Coordinate Geometry

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

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